Nonlinear functions — Part 2

Advanced Math · Math · 86 questions

2.1 a598add1-606e-453f-8b43-02083a30f5c7 · Medium

The value of a certain fishing boat was $\$ 24{,}000$ when it was purchased. Each year, the value of the fishing boat decreased by 20% of the previous year's value. Which equation represents this situation, where v is the value of the fishing boat, in dollars, and y is the number of years since the fishing boat was purchased?

A. $v = 0.8(24{,}000)^{y}$

B. $v = 1.2(24{,}000)^{y}$

C. $v = 24{,}000(0.8)^{y}$

D. $v = 24{,}000(1.2)^{y}$

2.2 969792b2-4ab7-4fe4-8d9e-e1a709101015 · Medium

The function $fw = 6w^2$ gives the area of a rectangle, $\text{in square feet } \text{ft}^2$, if its width is w ft and its length is 6 times its width. Which of the following is the best interpretation of $f14 = 1{,}176$?

A. If the width of the rectangle is 14 ft, then the area of the rectangle is $1{,}176$ ft^2.

B. If the width of the rectangle is 14 ft, then the length of the rectangle is $1{,}176$ ft.

C. If the width of the rectangle is $1{,}176$ ft, then the length of the rectangle is 14 ft.

D. If the width of the rectangle is $1{,}176$ ft, then the area of the rectangle is 14 ft^2.

2.3 025702-DC · Medium

$N(d) = 115(0.90)^d$

The function N defined above can be used to model the number of species of brachiopods at various ocean depths d, where d is in hundreds of meters. Which of the following does the model predict?

A. For every increase in depth by 1 meter, the number of brachiopod species decreases by 115.

B. For every increase in depth by 1 meter, the number of brachiopod species decreases by 10%.

C. For every increase in depth by 100 meters, the number of brachiopod species decreases by 115.

D. For every increase in depth by 100 meters, the number of brachiopod species decreases by 10%.

2.4 94ede15a-23c9-4c00-bcde-09c2bb206b79 · Medium

$$y = 7{,}4000.87^x$$

The given equation estimates the value, in dollars, of a certain piece of equipment x years after it was purchased. According to the given equation, what was the estimated value, in dollars, of the piece of equipment at the time it was purchased?

A. 740

B. 6,438

C. 7,400

D. 8,700

2.5 029384-DC · Medium

$f(x) = x^3 + 3x^2 - 6x - 1$

For the function f defined above, what is the value of $f(-1)$?

A. -11

B. -7

C. 7

D. 11

2.6 6951ac71-d64f-4a32-b84a-46f30f206692 · Medium

The function $ft = 60{,}0002^{\frac{t}{410}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

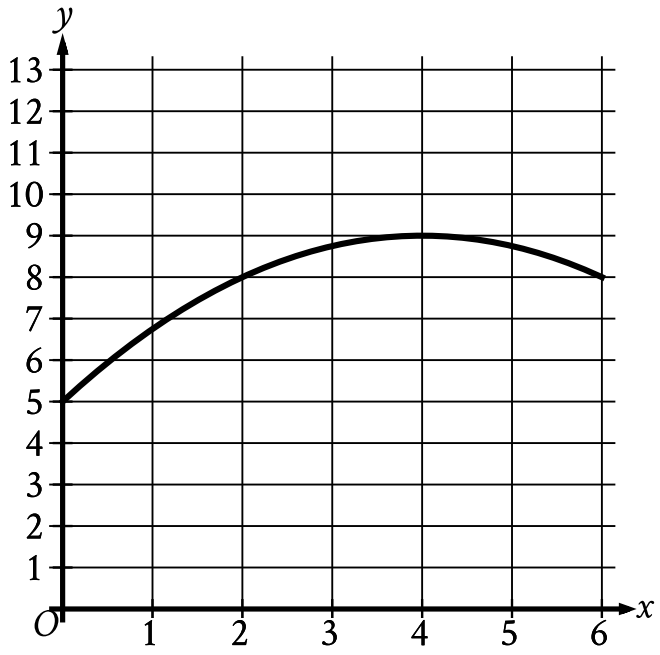
Student-produced response: _____

2.7 407267f8-d51c-4b0c-bd4e-fec1fe7b71c8 · Medium

The function h is defined by $hx = \frac{8}{5x+6}$. What is the value of $h2$?

Student-produced response: _____

2.8 d900a2e7-10e1-484e-a11a-0c2fe3bbb04f · Medium



- The parabola opens downward.
- The vertex is at point (4 comma 9).
- The parabola passes through the following points:
  - (0 comma 5)
  - (2 comma 8)
  - (4 comma 9)
  - (6 comma 8)

The graph models the number of active projects a company was working on x months after the end of November 2012, where $0 \leq x \leq 6$. According to the model, what is the predicted number of active projects the company was working on at the end of November 2012?

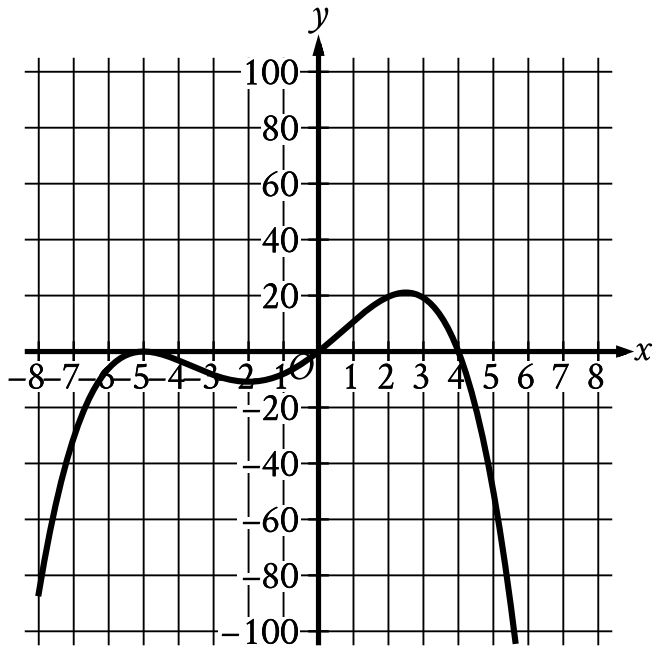
A. 0

B. 5

C. 8

D. 9

2.9 30c5bb08-3642-4126-a066-371cb194f2d1 · Medium



- In quadrant 3:
  - The curve rises sharply to touch the x axis at point (negative 5 comma 0).
  - The curve falls gradually to a relative minimum at point (negative 2 comma negative 11).
  - The curve rises gradually to cross both axes at the origin.
- In quadrant 1:
  - The curve rises gradually to a relative maximum at point (2.5 comma 21).
  - The curve falls sharply to cross the x axis at point (4 comma 0).
- In quadrant 4 the curve falls sharply.

Which of the following could be the equation of the graph shown in the *xy*-plane?

A. $y = -\frac{1}{10}xx - 4x + 5$

B. $y = -\frac{1}{10}xx - 4x + 5^2$

C. $y = -\frac{1}{10}xx - 5x + 4$

D. $y = -\frac{1}{10}xx - 5^2 x + 4$

2.10 025038-DC · Medium

A certain college had 3,000 students enrolled in 2015. The college predicts that after 2015, the number of students enrolled each year will be 2% less than the number of students enrolled the year before. Which of the following functions models the relationship between the number of students enrolled, $f(x)$, and the number of years after 2015, x ?

A. $f(x) = 0.02(3{,}000)^x$

B. $f(x) = 0.98(3{,}000)^x$

C. $f(x) = 3{,}000(0.02)^x$

D. $f(x) = 3{,}000(0.98)^x$

2.11 375d70d6-9387-4fb7-a51e-674183bfce7c · Medium

$$gx = x^2 + 55$$

What is the minimum value of the given function?

A. 0

B. 55

C. 110

D. 3,025

2.12 885439e9-2f0d-44e8-a3af-605c5924be71 · Medium

The function C defined by $Ct = 61.5^t$ gives the estimated number of clients at a certain business, where t is the number of weeks since the business opened. Which of the following is the best interpretation of the number 6 in this context?

A. The estimated number of clients when the business opened

B. The increase in the estimated number of clients each week

C. The number of weeks the business has been open

D. The percent increase in the estimated number of clients each week

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2.13 f41f1cb2-304e-4865-a112-f9de1e665a43 · Medium

$$y = 0.25x^2 - 7.5x + 90.25$$

The equation gives the estimated stock price y, in dollars, for a certain company x days after a new product launched, where $0 \leq x \leq 20$. Which statement is the best interpretation of $x, y = 1, 83$ in this context?

A. The company's estimated stock price increased $\$ 83$ every day after the new product launched.

B. The company's estimated stock price increased $\$ 1$ every 83 days after the new product launched.

C. 1 day after the new product launched, the company's estimated stock price is $\$ 83$.

D. 83 days after the new product launched, the company's estimated stock price is $\$ 1$.

2.14 ceec8040-0ddb-4726-b897-3baa00b904d2 · Medium

For the exponential function f, the value of $f0$ is c, where c is a constant. Of the following equations that define the function f, which equation shows the value of c as the coefficient or the base?

A. $fx = 221.5^{x+1}$

B. $fx = 331.5^{x}$

C. $fx = 49.51.5^{x-1}$

D. $fx = 74.251.5^{x-2}$

2.15 5186bb0a-09e9-4581-aae8-e440d02355fe · Medium

The function g is defined by $gx = 10^{5x-8}$. What is the value of x when gx is equal to $10{,}000$?

A. $\frac{11}{5}$

B. $\frac{12}{5}$

C. 3

D. 4

2.16 025805-DC · Medium

$D = 5{,}640(1.9)^t$

The equation above estimates the global data traffic D, in terabytes, for the year that is t years after 2010. What is the best interpretation of the number 5,640 in this context?

A. The estimated amount of increase of data traffic, in terabytes, each year

B. The estimated percent increase in the data traffic, in terabytes, each year

C. The estimated data traffic, in terabytes, for the year that is t years after 2010

D. The estimated data traffic, in terabytes, in 2010

2.17 585130f5-a566-4db1-900e-4f8d08cd67e6 · Medium

$$fx = \frac{a - 19}{x} + 5$$

In the given function f, a is a constant. The graph of function f in the xy-plane, where $y = fx$, is translated 3 units down and 4 units to the right to produce the graph of $y = gx$. Which equation defines function g?

A. $gx = \frac{a - 19}{x + 4} + 2$

B. $gx = \frac{a - 19}{x - 4} + 2$

C. $gx = \frac{a - 22}{x + 4} + 5$

D. $gx = \frac{a - 22}{x - 4} + 5$

2.18 b76e3ad3-f70a-40d5-b897-a3b5e368da81 · Medium

For the function q, the value of qx decreases by 45% for every increase in the value of x by 1. If $q0 = 14$, which equation defines q?

A. $qx = 0.5514^x$

B. $qx = 1.4514^x$

C. $qx = 140.55^x$

D. $qx = 141.45^x$

2.19 05eb7086-86be-4c43-bb55-070f148ab34a · Medium

An object is kicked from a platform. The equation $h = -4.9t^2 + 7t + 9$ represents this situation, where h is the height of the object above the ground, in meters, t seconds after it is kicked. Which number represents the height, in meters, from which the object was kicked?

A. 0

B. 4.9

C. 7

D. 9

2.20 e10027a7-cac8-4ca2-855a-ca3f00b55687 · Medium

The function $f(x) = 206(1.034)^x$ models the value, in dollars, of a certain bank account by the end of each year from 1957 through 1972, where x is the number of years after 1957. Which of the following is the best interpretation of "$f(5)$ is approximately equal to 243" in this context?

A. The value of the bank account is estimated to be approximately 5 dollars greater in 1962 than in 1957.

B. The value of the bank account is estimated to be approximately 243 dollars in 1962.

C. The value, in dollars, of the bank account is estimated to be approximately 5 times greater in 1962 than in 1957.

D. The value of the bank account is estimated to increase by approximately 243 dollars every 5 years between 1957 and 1972.

2.21 3f8c2f2d-abee-4738-aab3-11a77be98e29 · Medium

The product of a positive number x and the number that is 8 more than x is 180. What is the value of x?

A. 5

B. 10

C. 18

D. 36

2.22 98b110ad-3cb9-4925-831e-09572e226fda · Medium

The area A, in square centimeters, of a rectangular cutting board can be represented by the expression $ww + 9$, where w is the width, in centimeters, of the cutting board. Which expression represents the length, in centimeters, of the cutting board?

A. $ww + 9$

B. w

C. 9

D. $w + 9$

2.23 09291-DC · Medium

$$f(\theta) = -0.28(\theta - 27)^2 + 880$$

An engineer wanted to identify the best angle for a cooling fan in an engine in order to get the greatest airflow. The engineer discovered that the function above models the airflow $f(\theta)$, in cubic feet per minute, as a function of the angle of the fan θ, in degrees. According to the model, what angle, in degrees, gives the greatest airflow?

A. –0.28

B. 0.28

C. 27

D. 880

2.24 664fb796-9a6b-4891-a961-52cdf5f5b989 · Medium

x	fx
-1	10
0	14
1	20

For the quadratic function f, the table shows three values of x and their corresponding values of fx. Which equation defines f?

A. $fx = 3x^2 + 3x + 14$

B. $fx = 5x^2 + x + 14$

C. $fx = 9x^2 - x + 14$

D. $fx = x^2 + 5x + 14$

2.25 ca334282-00d9-4e64-9f61-edc6bc5d7646 · Medium

On April 1, there were 233 views of an advertisement posted on a website. Every 2 days after April 1, the number of views of the advertisement had increased by 70% of the number of views 2 days earlier. The function f gives the predicted number of views x days after April 1. Which equation defines f?

A. $fx = 2330.70^{\frac{x}{2}}$

B. $fx = 2330.70^{2x}$

C. $fx = 2331.70^{\frac{x}{2}}$

D. $fx = 2331.70^{2x}$

2.26 ce714e5d-1fb7-44e8-bfbc-55e6659cac72 · Medium

The area A, in square centimeters, of a rectangular painting can be represented by the expression $ww + 29$, where w is the width, in centimeters, of the painting. Which expression represents the length, in centimeters, of the painting?

A. w

B. 29

C. $w + 29$

D. $ww + 29$

2.27 f8a92cc0-95bf-4c16-8e6e-b31fa01f8b52 · Medium

x	y
1	11
2	19
3	a

The table shows three values of x and their corresponding values of y for the equation $y = 42^x + 3$. In the table, a is a constant. What is the value of a?

A. 67

B. 35

C. 32

D. 27

2.28 070617-DC · Medium

A rubber ball bounces upward one-half the height that it falls each time it hits the ground. If the ball was originally dropped from a distance of 20.0 feet above the ground, what was its maximum height above the ground, in feet, between the third and fourth time it hit the ground?

Student-produced response: _____

2.29 a8304c28-11ca-4f4b-b7bf-4379eab3e67e · Medium

The product of two positive integers is 546. If the first integer is 11 greater than twice the second integer, what is the smaller of the two integers?

A. 7

B. 14

C. 39

D. 78

2.30 4c9d9c55-af83-41a1-beaf-3cf5e261efd5 · Medium

A scientist initially measures $12{,}000$ bacteria in a growth medium. 4 hours later, the scientist measures $24{,}000$ bacteria. Assuming exponential growth, the formula $P = C2^{rt}$ gives the number of bacteria in the growth medium, where r and C are constants and P is the number of bacteria t hours after the initial measurement. What is the value of r?

A. $\frac{1}{12{,}000}$

B. $\frac{1}{4}$

C. 4

D. $12{,}000$

2.31 1b3396a5-95c8-4241-a154-8fc5c60e081b · Medium

$$fx = 3{,}0000.75^{x}$$

A conservation scientist implemented a program to reduce the population of a certain species in an area. The given function estimates this species' population x years after 2008, where $x \leq 8$. Which of the following is the best interpretation of $3{,}000$ in this context?

A. The estimated percent decrease in the population for this species and area every 8 years after 2008

B. The estimated percent decrease in the population for this species and area each year after 2008

C. The estimated population for this species and area 8 years after 2008

D. The estimated initial population for this species and area in 2008

2.32 61c05d04-1600-452b-9723-85d1636b8f5a · Medium

$$px + 57 = x^{2}$$

The given equation relates the value of x and its corresponding value of px for the function p. What is the minimum value of the function p?

A. $-3{,}249$

B. -57

C. 57

D. $3{,}249$

2.33 028933-DC · Medium

From 2005 through 2014, the number of music CDs sold in the United States declined each year by approximately 15% of the number sold the preceding year. In 2005, approximately 600 million CDs were sold in the United States. Of the following, which best models C, the number of millions of CDs sold in the United States, t years after 2005?

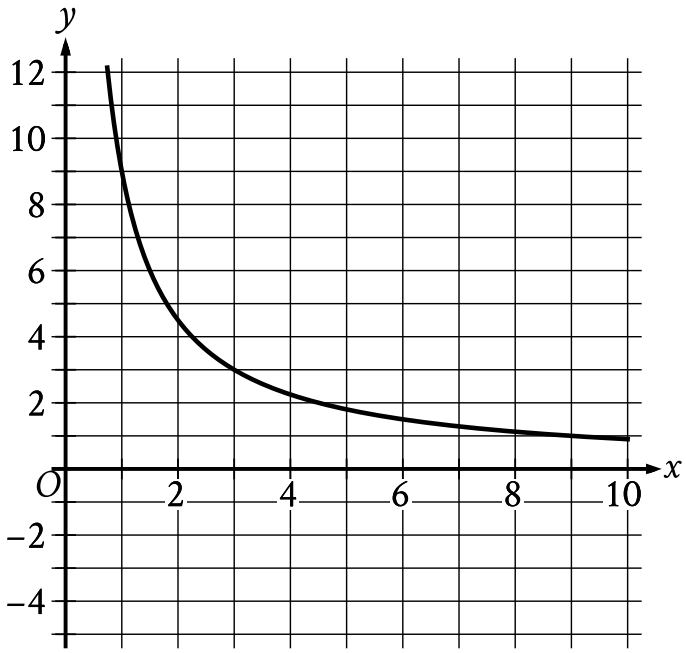
A. $C = 600(0.15)^t$

B. $C = 600(0.85)^t$

C. $C = 600(1.15)^t$

D. $C = 600(1.85)^t$

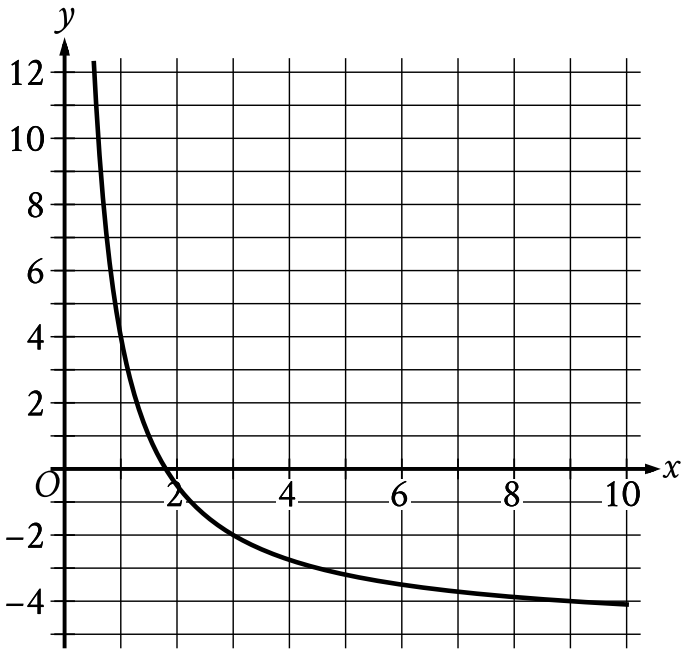
2.34 9025546d-7856-480c-9be8-6cbd532f45ee · Medium



- Moving from left to right:
  - The curve is shown in quadrant 1.
  - The curve trends down sharply to x equals 3.
  - The curve then trends down gradually.
- As x increases, the curve approaches the line y equals 0.
- The curve passes through the following points:
  - (2 comma nine halves)
  - (3 comma 3)
  - (6 comma three halves)

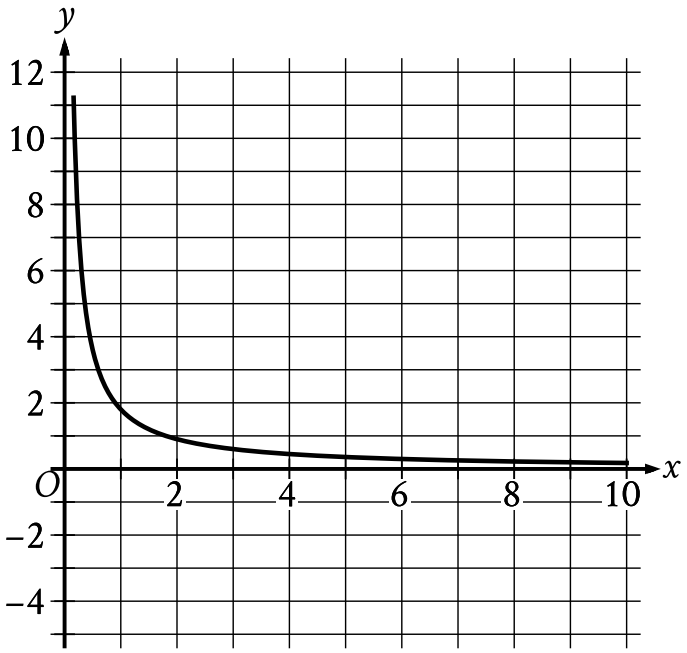
The graph of the rational function f is shown, where $y = f(x)$ and $x \geq 0$. Which of the following is the graph of $y = f(x) + 5$, where $x \geq 0$?

A.



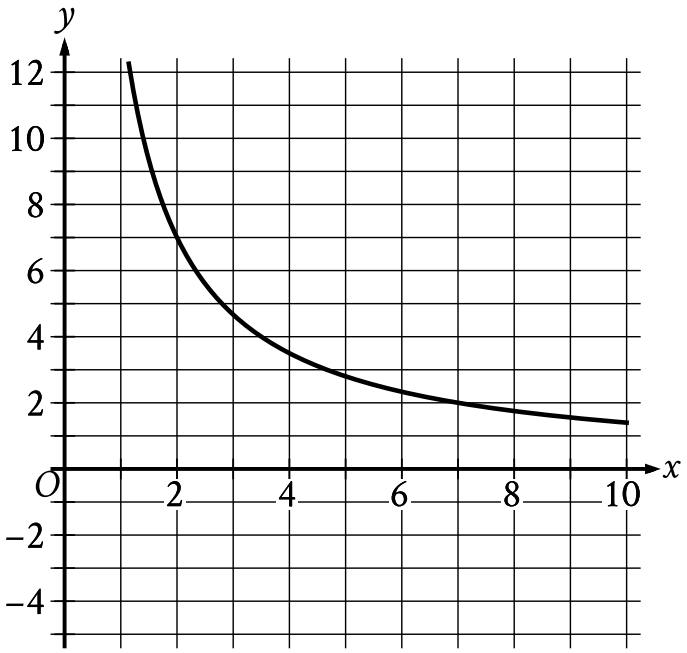
- Moving from left to right:
  - The curve passes from quadrant 1 to quadrant 4.
  - The curve trends down sharply to x equals 3.
  - The curve then trends down gradually.
- As x increases, the curve approaches the line y equals negative 5.
- The curve passes through the following points:
  - (2 comma negative one half)
  - (3 comma negative 2)
  - (6 comma negative seven halves)

B.



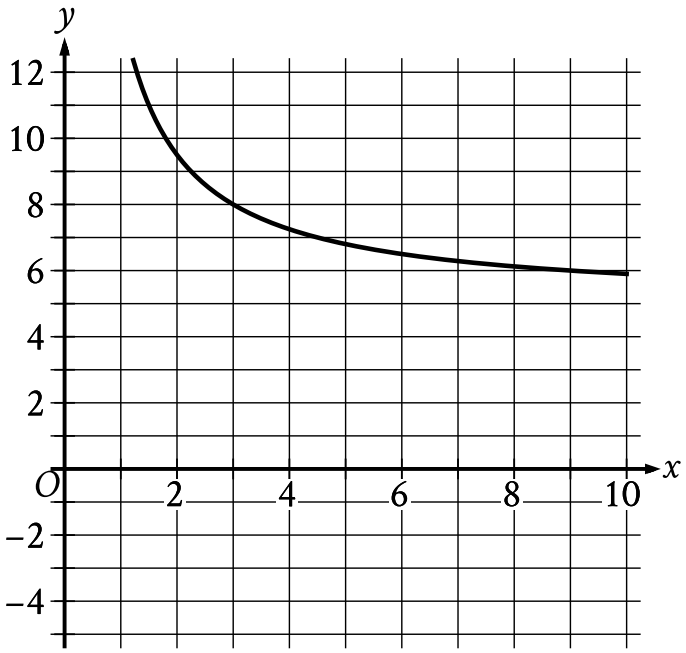
- Moving from left to right:
  - The curve is shown in quadrant 1.
  - The curve trends down sharply to x equals 1.
  - The curve then trends down gradually.
- As x increases, the curve approaches the line y equals 0.
- The curve passes through the following points:
  - (2 comma nine tenths)
  - (3 comma three fifths)
  - (6 comma three tenths)

C.



- Moving from left to right:
  - The curve is shown in quadrant 1.
  - The curve trends down sharply to x equals 2.
  - The curve then trends down gradually.
- As x increases, the curve approaches the line y equals 0.
- The curve passes through the following points:
  - (2 comma 7)
  - (3 comma StartFraction 14 Over 3 EndFraction)
  - (6 comma seven thirds)

D.



- Moving from left to right:
  - The curve is shown in quadrant 1.
  - The curve trends down sharply to x equals 3.
  - The curve then trends down gradually.
- As x increases, the curve approaches the line y equals 5.
- The curve passes through the following points:
  - (2 comma StartFraction 19 Over 2 EndFraction)
  - (3 comma 8)
  - (6 comma StartFraction 13 Over 2 EndFraction)

2.35 46f2fda7-d91a-4059-9f4c-9f57b65cf702 · Medium

The function $ft = 40{,}0002^{\frac{t}{790}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

A. 2

B. 790

C. $1{,}580$

D. $40{,}000$

2.36 853afe28-d4da-47b1-abf4-817c4a543239 · Medium

x	24	30	32
fx	-8	-8	8

For the quadratic function f, the table shows three values of x and their corresponding values of fx. Which equation defines f?

A. $f(x) = (x - 24)(x - 30) + 4$

B. $f(x) = (x - 24)(x - 30) - 8$

C. $f(x) = (x - 8)(x - 32) + 32$

D. $f(x) = (x - 8)(x - 32) - 32$

2.37 8472776c-becb-4a82-bf1b-aa29eb25da77 · Medium

The equation $Et = 51.8^t$ gives the estimated number of employees at a restaurant, where t is the number of years since the restaurant opened. Which of the following is the best interpretation of the number 5 in this context?

A. The estimated number of employees when the restaurant opened

B. The increase in the estimated number of employees each year

C. The number of years the restaurant has been open

D. The percent increase in the estimated number of employees each year

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2.38 85dc4466-29e1-4dd1-8c9b-cf1c1f0d641f · Medium

The function f is defined by $fx = 10x^2 - 32x - 152$. What is the value of $f0$?

A. -152

B. -32

C. 0

D. 10

2.39 d73707f3-c5fe-430b-8799-06ef95532995 · Medium

Immanuel purchased a certain rare coin on January 1. The function $f(x) = 651.03^x$, where $0 \leq x \leq 10$, gives the predicted value, in dollars, of the rare coin x years after Immanuel purchased it. What is the best interpretation of the statement "$f8$ is approximately equal to 82" in this context?

A. When the rare coin's predicted value is approximately 82 dollars, it is 8% greater than the predicted value, in dollars, on January 1 of the previous year.

B. When the rare coin's predicted value is approximately 82 dollars, it is 8 times the predicted value, in dollars, on January 1 of the previous year.

C. From the day Immanuel purchased the rare coin to 8 years after Immanuel purchased the coin, its predicted value increased by a total of approximately 82 dollars.

D. 8 years after Immanuel purchased the rare coin, its predicted value is approximately 82 dollars.

2.40 abfd9618-c65d-4fd8-9f89-254e3dfa9529 · Medium

A rectangle has a length that is 15 times its width. The function $y = 15ww$ represents this situation, where y is the area, in square feet, of the rectangle and $y > 0$. Which of the following is the best interpretation of $15w$ in this context?

A. The length of the rectangle, in feet

B. The area of the rectangle, in square feet

C. The difference between the length and the width of the rectangle, in feet

D. The width of the rectangle, in feet

2.41 029475-DC · Medium

In the xy-plane, the y-coordinate of the y-intercept of the graph of the function f is c. Which of the following must be equal to c ?

A. $f(0)$

B. $f(1)$

C. $f(2)$

D. $f(3)$

2.42 ccdb450a-31b3-42ec-9532-4859dd3972c3 · Medium

$$fx = x^2 - 18x - 360$$

If the given function f is graphed in the xy-plane, where $y = fx$, what is an x-intercept of the graph?

A. $-12, 0$

B. $-30, 0$

C. $-360, 0$

D. $12, 0$

2.43 d0812cf6-f4e9-4df1-9620-878d9a37d592 · Medium

An egg is thrown from a rooftop. The equation $h = -4.9t^2 + 9t + 18$ represents this situation, where h is the height of the egg above the ground, in meters, t seconds after it is thrown. According to the equation, what is the height, in meters, from which the egg was thrown?

Student-produced response: _____

2.44 b43f007e-8a7c-48b8-8a0a-eeb0f65e3faf · Medium

A model predicts that the population of Bergen was $15{,}000$ in 2005. The model also predicts that each year for the next 5 years, the population p increased by 4% of the previous year's population. Which equation best represents this model, where x is the number of years after 2005, for $x \leq 5$?

A. $p = 0.9615{,}000^x$

B. $p = 1.0415{,}000^x$

C. $p = 15{,}0000.96^x$

D. $p = 15{,}0001.04^x$

2.45 0ef7164f-e18c-4aa3-999c-ece225b93166 · Medium

The function f is defined by $fx = 4x^{-1}$. What is the value of $f21$?

A. -84

B. $\frac{1}{84}$

C. $\frac{4}{21}$

D. $\frac{21}{4}$

2.46 45800f0b-d83c-4237-a7a2-1f84edd039a1 · Medium

A physics class is planning an experiment about a toy rocket. The equation $y = -16x - 5.6^2 + 502$ gives the estimated height y, in feet, of the toy rocket x seconds after it is launched into the air. Which of the following is the best interpretation of the vertex of the graph of the equation in the xy-plane?

A. This toy rocket reaches an estimated maximum height of 502 feet 16 seconds after it is launched into the air.

B. This toy rocket reaches an estimated maximum height of 502 feet 5.6 seconds after it is launched into the air.

C. This toy rocket reaches an estimated maximum height of 16 feet 502 seconds after it is launched into the air.

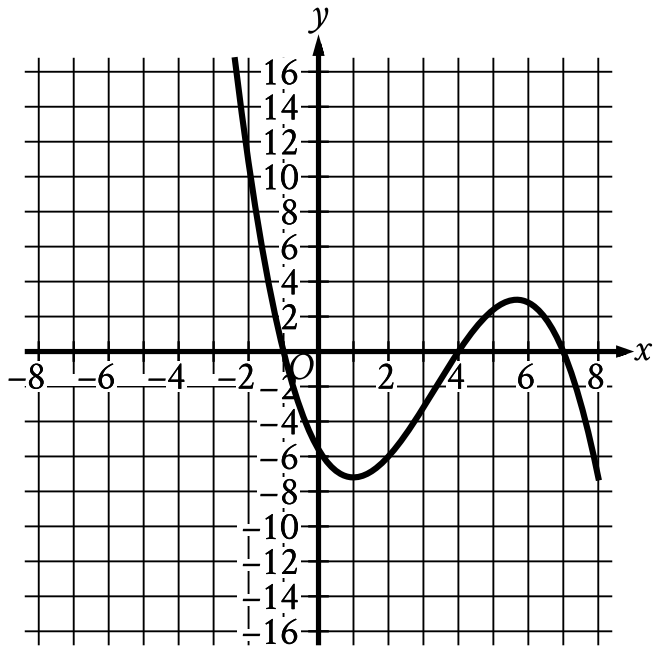
D. This toy rocket reaches an estimated maximum height of 5.6 feet 502 seconds after it is launched into the air.

2.47 59addf61-9f46-4c0d-9dce-ebce1fa135b7 · Medium

The function f is defined by $f(x) = 5\frac{1}{4} - x^2 + \frac{11}{4}$. What is the value of $f\left(\frac{1}{4}\right)$?

Student-produced response: _____

2.48 d7821fa6-7c23-42a3-8896-9b2a32bffb92 · Medium



- In quadrant 2 the curve falls sharply to cross the x axis at point (negative 1 comma 0).
- In quadrant 3 the curve falls sharply to cross the y axis at point (0 comma negative 5.5).
- In quadrant 4:
  - The curve falls gradually to a relative minimum at point (1 comma negative 7).
  - The curve rises sharply to cross the x axis at point (4 comma 0).
- In quadrant 1:
  - The curve rises sharply to a relative maximum at point (5.5 comma 3).
  - The curve falls sharply to cross the x axis at point (7 comma 0).
- In quadrant 4 the curve falls sharply.

The graph of $y = fx$ is shown, where the function f is defined by $fx = ax^3 + bx^2 + cx + d$ and a, b, c, and d are constants. For how many values of x does $fx = 0$?

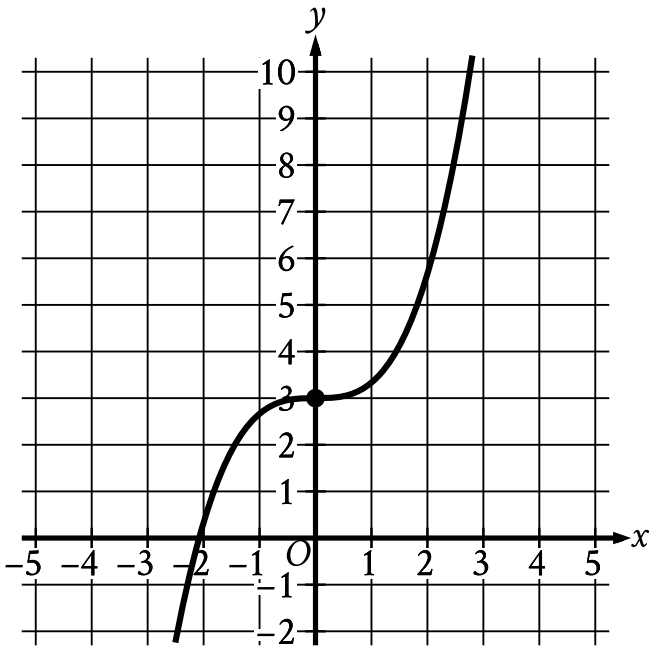
A. One

B. Two

C. Three

D. Four

2.49 eb535e3d-3c71-4caf-883c-c6a73134317d · Medium



- Moving from left to right, the curve rises from quadrant 3 to quadrant 2 to quadrant 1.
- The curve passes through the following points:
  - (negative 1 comma eight thirds)
  - (0 comma 3)
  - (1 comma StartFraction 10 Over 3 EndFraction)

The graph of $y = fx$ is shown. What is the y-intercept of $y = fx + 5$?

A. $0, 8$

B. $0, 5$

C. $0, 3$

D. $0, 0$

2.50 016491-DC · Medium

The height, in feet, of an object x seconds after it is thrown straight up in the air can be modeled by the function $h(x) = -16x^2 + 20x + 5$. Based on the model, which of the following statements best interprets the equation $h(1.4) = 1.64$?

A. The height of the object 1.4 seconds after being thrown straight up in the air is 1.64 feet.

B. The height of the object 1.64 seconds after being thrown straight up in the air is 1.4 feet.

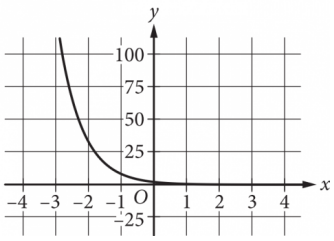
C. The height of the object 1.64 seconds after being thrown straight up in the air is approximately 1.4 times as great as its initial height.

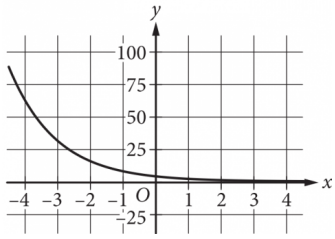
D. The speed of the object 1.4 seconds after being thrown straight up in the air is approximately 1.64 feet per second.

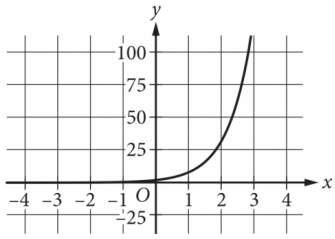
2.51 027803-DC · Medium

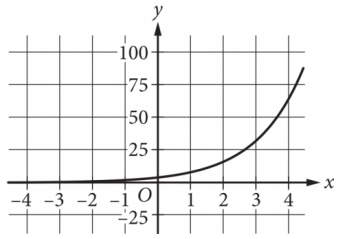
$y = 4(2^x)$

Which of the following is the graph in the xy-plane of the given equation?

A. 

B. 

C. 

D. 

2.52 ecc55958-e485-4d44-84c5-3a70884d1fb5 · Medium

A rectangle has a length of x units and a width of $x - 15$ units. If the rectangle has an area of 76 square units, what is the value of x?

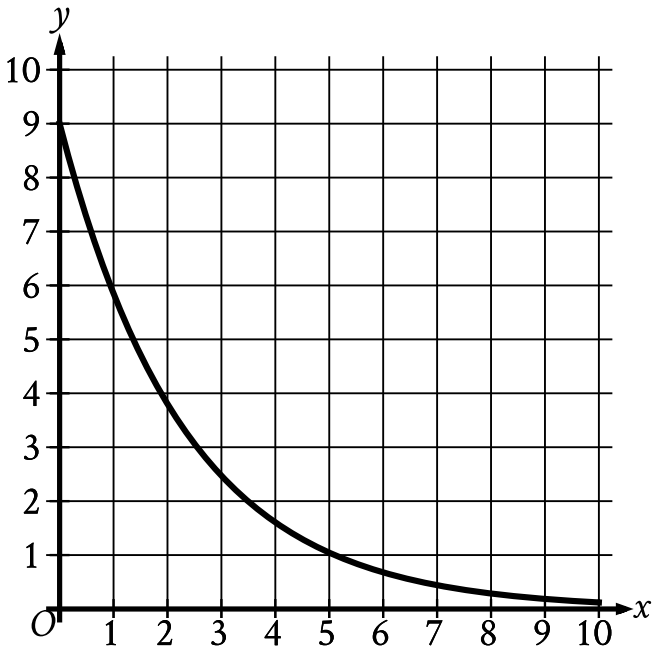
A. 4

B. 19

C. 23

D. 76

2.53 433ee68c-fedc-4334-84d7-4032b35bf0e2 · Medium



- Moving from left to right:
  - The curve is in quadrant 1.
  - From point (0 comma 9), the curve trends down sharply.
  - From point (5 comma 1), the curve trends down gradually.

The graph gives the estimated number of catalogs y, in thousands, a company sent to its customers at the end of each year, where x represents the number of years since the end of 1992, where $0 \le x \le 10$. Which statement is the best interpretation of the y-intercept in this context?

A. The estimated total number of catalogs the company sent to its customers during the first 10 years was $9{,}000$.

B. The estimated total number of catalogs the company sent to its customers from the end of 1992 to the end of 2002 was 90.

C. The estimated number of catalogs the company sent to its customers at the end of 1992 was 9.

D. The estimated number of catalogs the company sent to its customers at the end of 1992 was $9{,}000$.

2.54 017018-DC · Medium

x	f(x)
0	5
1	$\frac{5}{2}$
2	$\frac{5}{4}$
3	$\frac{5}{8}$

The table above gives the values of the function f for some values of x. Which of the following equations could define f ?

A. $f(x) = 5(2^{x+1})$

B. $f(x) = 5(2^{x})$

C. $f(x) = 5(2^{-(x+1)})$

D. $f(x) = 5(2^{-x})$

2.55 20b0ac3b-55d7-436b-be64-cb9086cccb62 · Medium

The function *f* is defined by $fx = 8x^3 + 4$. What is the value of $f2$?

Student-produced response: _____

2.56 044c0858-a7b2-4913-8535-7de0c2051f19 · Medium

$$ft = 5000.5^{\frac{t}{12}}$$

The function f models the intensity of an X-ray beam, in number of particles in the X-ray beam, t millimeters below the surface of a sample of iron. According to the model, what is the estimated number of particles in the X-ray beam when it is at the surface of the sample of iron?

A. 500

B. 12

C. 5

D. 2

2.57 b026eab4-9126-4926-ba7f-aeb53aa41e01 · Medium

$$fx = x^5 + 9x + 17$$

For the given function f, the graph of $y = fx$ in the xy-plane passes through the point $0, b$, where b is a constant. What is the value of b?

Student-produced response: _____

2.58 dbd2580c-f8fe-48de-b7f0-838c5fe91f7c · Medium

Time (years)	Total amount (dollars)
0	670.00
1	674.02
2	678.06

Sara opened a savings account at a bank. The table shows the exponential relationship between the time t, in years, since Sara opened the account and the total amount d, in dollars, in the account. If Sara made no additional deposits or withdrawals, which of the following equations best represents the relationship between t and d?

A. $d = 0.0061 + 670^t$

B. $d = 6701 + 0.006^t$

C. $d = 1 + 0.006^t$

D. $d = 1 + 670^t$

2.59 ff8903c3-0d12-4c8b-8aef-3372dfbd9793 · Medium

$$ht = -16t^2 + b$$

The function h estimates an object's height, in feet, above the ground t seconds after the object is dropped, where b is a constant. The function estimates that the object is $3{,}364$ feet above the ground when it is dropped at $t = 0$. Approximately how many seconds after being dropped does the function estimate the object will hit the ground?

A. 7.25

B. 14.50

C. 105.13

D. 210.25

2.60 029350-DC · Medium

$S(n) = 38{,}000a^n$

The function S above models the annual salary, in dollars, of an employee n years after starting a job, where a is a constant. If the employee's salary increases by 4% each year, what is the value of a ?

A. 0.04

B. 0.4

C. 1.04

D. 1.4

2.61 025772-DC · Medium

According to Moore's law, the number of transistors included on microprocessors doubles every 2 years. In 1985, a microprocessor was introduced that had 275,000 transistors. Based on this information, in which of the following years does Moore's law estimate the number of transistors to reach 1.1 million?

A. 1987

B. 1989

C. 1991

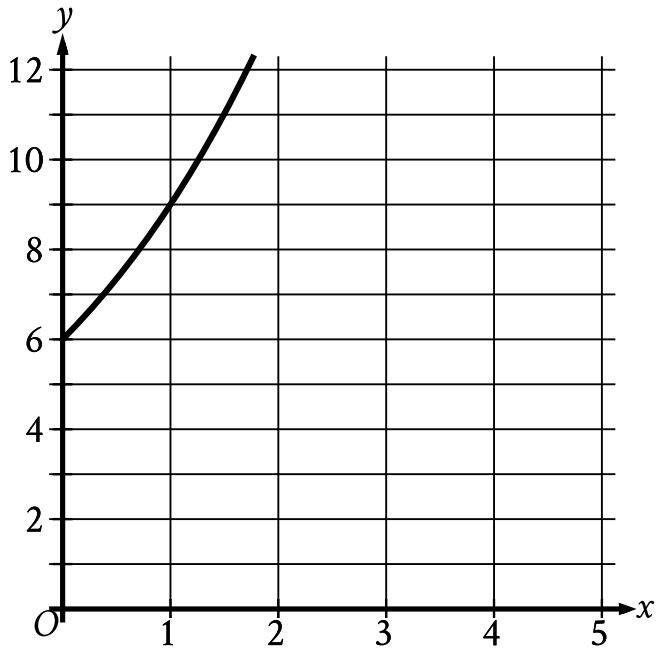
D. 1994

2.62 9accecc3-e743-4bfa-9e4f-235283d99abd · Medium

The exponential function g is defined by $gx = 19 \cdot a^x$, where a is a positive constant. If $g3 = 2{,}375$, what is the value of $g4$?

Student-produced response: _____

2.63 8fbdc9d2-8c5f-4da3-98db-20599f2fff76 · Medium



- Moving from left to right:
  - The curve passes through quadrant 1.
  - The curve trends up sharply.
- The curve passes through the following points:
  - (0 comma 6)
  - (1 comma 9.0)

The graph gives the estimated population y, in thousands, of a town x years since 2003, where $0 \leq x \leq 5$. Which of the following best describes the increase in the estimated population from $x = 0$ to $x = 1$?

A. The estimated population at $x = 1$ is 0.5 times the estimated population at $x = 0$.

B. The estimated population at $x = 1$ is 1.5 times the estimated population at $x = 0$.

C. The estimated population at $x = 1$ is 2.5 times the estimated population at $x = 0$.

D. The estimated population at $x = 1$ is 3.5 times the estimated population at $x = 0$.

2.64 028940-DC · Medium

$f(x) = (x+4)(x-1)(2x-3)$

The function f is defined above. Which of the following is NOT an x-intercept of the graph of the function in the xy-plane?

A. $(-4, 0)$

B. $\left(-\frac{2}{3}, 0\right)$

C. $(1, 0)$

D. $\left(\frac{3}{2}, 0\right)$

2.65 045551-DC · Medium

An object's kinetic energy, in joules, is equal to the product of one-half the object's mass, in kilograms, and the square of the object's speed, in meters per second. What is the speed, in meters per second, of an object with a mass of 4 kilograms and kinetic energy of 18 joules?

A. 3

B. 6

C. 9

D. 36

2.66 021775-DC · Medium

A rectangular volleyball court has an area of 162 square meters. If the length of the court is twice the width, what is the width of the court, in meters?

A. 9

B. 18

C. 27

D. 54

2.67 fdf648c1-f590-4c15-b580-2bbbdeb7df2f · Medium

The area of a triangle is 270 square centimeters. The length of the base of the triangle is 12 centimeters greater than the height of the triangle. What is the height, in centimeters, of the triangle?

A. 15

B. 18

C. 30

D. 36

2.68 7888cfcc-ba3d-422a-be02-86652b46cb9d · Medium

$$q(x) = 32(2^x)$$

Which table gives three values of x and their corresponding values of $q(x)$ for function q?

A.

x	-1	0	1
$q(x)$	-64	0	64

B.

x	-1	0	1
$q(x)$	$\frac{1}{16}$	2	64

C.

x	-1	0	1
$q(x)$	$\frac{1}{16}$	32	64

D.

x	-1	0	1
$q(x)$	16	32	64

2.69 2a0b1f88-5e2b-441c-84cb-141f8592d02a · Medium

x	hx
0	1.23
2	1.54
4	1.94

The table shows the exponential relationship between the number of years, x, since Hana started training in pole vault, and the estimated height hx, in meters, of her best pole vault for that year. Which of the following functions best represents this relationship, where $x \leq 4$?

A. $hx = 1.120.23^x$

B. $hx = 1.121.23^x$

C. $hx = 1.230.12^x$

D. $hx = 1.231.12^x$

2.70 f9da1b3c-c8a6-40ae-b912-90cb932f1c07 · Medium

x	gx
-1	25
0	1
1	$\frac{1}{25}$
2	$\frac{1}{625}$

For the exponential function g, the table shows four values of x and their corresponding values of gx. Which equation defines g?

A. $gx = -25^x$

B. $gx = -\frac{1}{25}^x$

C. $gx = 25^x$

D. $gx = \frac{1}{25}^x$

2.71 ee6062be-ade1-48a3-b556-2cca3b7b45e3 · Hard

$$rt = 63t - 2t^2$$

The function r is defined by the given equation. The function s is defined by $st = rt + 1$. Which expression represents the maximum value of st?

A. $1 + \frac{63}{2}^2$

B. $1 + 2\frac{63}{4}^2$

C. $1 - 2\frac{63}{4}^2$

D. $1 - \frac{63}{2}^2$

2.72 7fc3b869-8f5c-4c0b-a28b-acbe8ea76ddf · Hard

$$fx = 9(4)^x$$

The function f is defined by the given equation. If $g(x) = f(x + 2)$, which of the following equations defines the function g?

A. $gx = 184^x$

B. $gx = 1444^x$

C. $gx = 188^x$

D. $gx = 8116^x$

2.73 04172-DC · Hard

Let the function p be defined as $p(x) = \frac{(x-c)^2 + 160}{2c}$, where c is a constant. If $p(c) = 10$, what is the value of $p(12)$?

A. 10.00

B. 10.25

C. 10.75

D. 11.00

2.74 baa55cdd-6e64-416a-8f18-d484e43741e8 · Hard

$$ft = 8{,}0000.65^{t}$$

The given function f models the number of coupons a company sent to their customers at the end of each year, where t represents the number of years since the end of 1998, and $0 \le t \le 5$. If $y = ft$ is graphed in the ty-plane, which of the following is the best interpretation of the y-intercept of the graph in this context?

A. The minimum estimated number of coupons the company sent to their customers during the 5 years was $1{,}428$.

B. The minimum estimated number of coupons the company sent to their customers during the 5 years was $8{,}000$.

C. The estimated number of coupons the company sent to their customers at the end of 1998 was $1{,}428$.

D. The estimated number of coupons the company sent to their customers at the end of 1998 was $8{,}000$.

2.75 fb591f40-b171-470c-ab62-48c9c0ee004b · Hard

$$pt = 90{,}0001.06^{t}$$

The given function p models the population of Lowell t years after a census. Which of the following functions best models the population of Lowell m months after the census?

A. $rm = \frac{90{,}000}{12}1.06^{m}$

B. $rm = 90{,}000\frac{1.06}{12}^{m}$

C. $rm = 90{,}000\frac{1.06}{12}^{\frac{m}{12}}$

D. $rm = 90{,}0001.06^{\frac{m}{12}}$

2.76 2da8e72d-a24c-49d7-a9ed-5f47b4fecf08 · Hard

The quadratic function g models the depth, in meters, below the surface of the water of a seal t minutes after the seal entered the water during a dive. The function estimates that the seal reached its maximum depth of 302.4 meters 6 minutes after it entered the water and then reached the surface of the water 12 minutes after it entered the water. Based on the function, what was the estimated depth, to the nearest meter, of the seal 10 minutes after it entered the water?

Student-produced response: _____

2.77 19eed232-6ab6-4e1d-8586-46e6d6357a09 · Hard

The first term of a sequence is 9. Each term after the first is 4 times the preceding term. If w represents the nth term of the sequence, which equation gives w in terms of n?

A. $w = 49^n$

B. $w = 49^{n-1}$

C. $w = 94^n$

D. $w = 94^{n-1}$

2.78 e6248b96-6cdd-48a4-a20b-dc4b62d59d56 · Hard

$$Pt = 2901.04^{\frac{4}{6}t}$$

The function P models the population, in thousands, of a certain city t years after 2005. According to the model, the population is predicted to increase by $n\%$ every 18 months. What is the value of n?

A. 0.38

B. 1.04

C. 4

D. 6

2.79 72f977c3-5379-42b1-947f-d9fc4dbc27fb · Hard

$$fx = 59 - 2x$$

The function f is defined by the given equation. For which of the following values of k does $fk = 3k$?

A. $\frac{59}{5}$

B. $\frac{59}{2}$

C. $\frac{177}{5}$

D. 59

2.80 35b45e44-2494-439f-a174-5c714299c55c · Hard

The function f is defined by $f(x) = a^x + b$, where a and b are constants. In the xy-plane, the graph of $y = fx$ has an x-intercept at $2, 0$ and a y-intercept at $0, -323$. What is the value of b?

Student-produced response: _____

2.81 59249867-1d27-418b-aac8-59cddb26069b · Hard

$$fx = x + 7^2 + 4$$

The function f is defined by the given equation. For what value of x does fx reach its minimum?

Student-produced response: _____

2.82 65743096-f9d1-4ab0-9a10-3acd3ea46114 · Hard

$$fx = x - 10x + 13$$

The function f is defined by the given equation. For what value of x does fx reach its minimum?

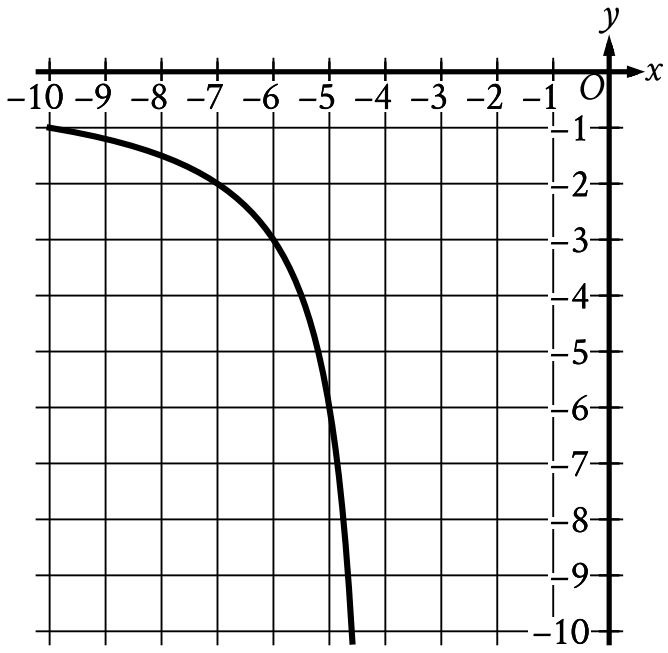
A. -130

B. -13

C. $-\frac{23}{2}$

D. $-\frac{3}{2}$

2.83 5af7e994-3850-4c5c-987d-1213c4f6baf8 · Hard



- Moving from left to right:
  - The curve is in quadrant 3.
  - The curve trends down sharply.
- As x increases, the curve approaches the line x equals negative 4.
- As x decreases, the curve approaches the line y equals 0.
- The curve passes through the following points:
  - (negative 7 comma negative 2)
  - (negative 6 comma negative 3)
  - (negative 5 comma negative 6)

The rational function f is defined by an equation in the form $fx = \frac{a}{x+b}$, where a and b are constants. The partial graph of $y = fx$ is shown. If $gx = fx + 4$, which equation could define function g?

A. $gx = \frac{6}{x}$

B. $gx = \frac{6}{x+4}$

C. $gx = \frac{6}{x+8}$

D. $gx = \frac{6(x+4)}{x+4}$

2.84 c1e6cb10-5989-4a1e-a246-78972ed7bf5a · Hard

$$f(x) = ax^2 + 4x + c$$

In the given quadratic function, a and c are constants. The graph of $y = f(x)$ in the xy-plane is a parabola that opens upward and has a vertex at the point (h, k), where h and k are constants. If $k < 0$ and $f(-9) = f(3)$, which of the following must be true?

I. $c < 0$
II. $a \geq 1$

A. I only

B. II only

C. I and II

D. Neither I nor II

2.85 f78a967a-96bf-469b-bc93-679f3d5dac92 · Hard

The function f is defined by $f(x) = 56(0.19)^x$. For any positive integer n, the value of $f(n)$ is $p\%$ less than the value of $f(n - 1)$. What is the value of p?

A. 19

B. 44

C. 56

D. 81

2.86 1010a838-bf8c-4e58-8c76-a23fa7c4ee2b · Hard

$$f(x) = 4x^2 - 50x + 126$$

The given equation defines the function f. For what value of x does $f(x)$ reach its minimum?

Student-produced response: _____
